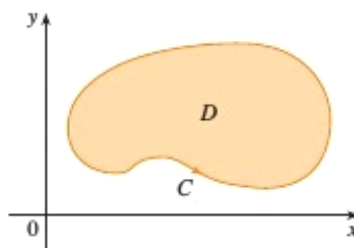


16.4 Green's Theorem

Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C .



Green's Theorem Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_C P \, dx + Q \, dy$$

$$\int_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Green's Theorem gives the following formulas for the area of D :

$$A = \oint_C x \, dy = -\oint_C y \, dx = \frac{1}{2} \oint_C x \, dy - y \, dx$$

EXAMPLE Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

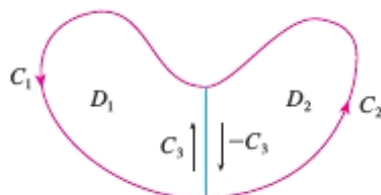
$$A = \frac{1}{2} \int_C x \, dy - y \, dx$$

$$= \frac{1}{2} \int_0^{2\pi} (a \cos t)(b \cos t) \, dt - (b \sin t)(-a \sin t) \, dt$$

$$= \frac{ab}{2} \int_0^{2\pi} dt = \pi ab$$

$$\begin{cases} P(x, y) = -\frac{1}{2}y \\ Q(x, y) = \frac{1}{2}x \end{cases}$$

Extended Versions of Green's Theorem



$$\int_{C_1 \cup C_2} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Green's Theorem for $D = D_1 \cup D_2$, since its boundary is $C = C_1 \cup C_2$.

16.5 Curl and Divergence

Curl

If $F = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R all exist, then the curl of F is the vector field on $\mathbb{R}^3$ defined by

$$\text{curl } F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

easiest way to remember $\text{curl } F$

$$\text{curl } F = \nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Theorem If f is a function of three variables that has continuous second-order partial derivatives, then

$$\text{curl}(\nabla f) = 0$$

Since a conservative vector field is one for which $F = \nabla f$, Theorem can be rephrased as follows:

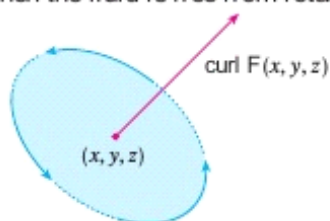
If F is conservative, then $\text{curl } F = 0$.

This gives a way of verifying that a vector field is not conservative.

Theorem If F is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and $\text{curl } F = 0$, then F is a conservative vector field.

The reason for the name curl is that the curl vector is associated with rotations.

when F represents the velocity field in fluid flow : Particles near (x, y, z) in the fluid tend to rotate about the axis that points in the direction of $\text{curl } F(x, y, z)$, and the length of this curl vector is a measure of how quickly the particles move around the axis (see Figure 1). If $\text{curl } F = 0$ at a point P , then the fluid is free from rotations at P and F is called irrotational at P .



Divergence

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$, and $\partial R/\partial z$ exist, then the divergence of $\mathbf{F}$ is the function of three variables defined by

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$\operatorname{curl} \mathbf{F}$ is a vector field but $\operatorname{div} \mathbf{F}$ is a scalar field.

the divergence of $\mathbf{F}$ written symbolically

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

as the dot product of ∇ and $\mathbf{F}$
 $\nabla = (\partial/\partial x) \mathbf{i} + (\partial/\partial y) \mathbf{j} + (\partial/\partial z) \mathbf{k}$

If $\mathbf{F}$ is a vector field on $\mathbb{R}^3$, then $\operatorname{curl} \mathbf{F}$ is also a vector field on $\mathbb{R}^3$. As such, compute its divergence. The next theorem shows that the result is 0.

Theorem If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then

$$\operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

!!! the reason for the name *divergence* can be understood in the context of fluid flow. If $\mathbf{F}(x, y, z)$ is the velocity of a fluid (or gas), then $\operatorname{div} \mathbf{F}(x, y, z)$ represents the net rate of change (with respect to time) of the mass of fluid (or gas) flowing from the point (x, y, z) per unit volume. In other words, $\operatorname{div} \mathbf{F}(x, y, z)$ measures the tendency of the fluid to diverge from the point (x, y, z) . If $\operatorname{div} \mathbf{F} = 0$, then $\mathbf{F}$ is said to be **incompressible**.